SMART BLINDS / INTEGRATED CONTROLLER

196 circuit features. USB-C charging, protected pack input, radio, motor and local controls.

Sheet labels share only the explicitly named global nets. Open a sheet to inspect every part.

USB-C input

33 parts / 113 pin identities

File: 01-usb.kicad_sch

3S charger

36 parts / 101 pin identities

File: 02-charger.kicad_sch

Logic supply

12 parts / 31 pin identities

File: 03-logic.kicad_sch

Motor bridge

14 parts / 43 pin identities

File: 04-motor.kicad_sch

Motor supervision

14 parts / 49 pin identities

File: 05-supervision.kicad_sch

Regenerative clamp

38 parts / 105 pin identities

File: 06-regen.kicad_sch

Radio and USB data

15 parts / 68 pin identities

File: 07-radio.kicad_sch

Encoder input

10 parts / 29 pin identities

File: 08-encoder-input.kicad_sch

Buttons and home input

11 parts / 30 pin identities

File: 09-local-input.kicad_sch

Power and motor permission

13 parts / 40 pin identities

File: 10-interfaces.kicad_sch

Captured circuit and native netlist are checked pin for pin. No routed or powered hardware acceptance.

Unrouted circuit; not for fabrication

Sheet: /
File: smart-blinds-main.kicad_sch

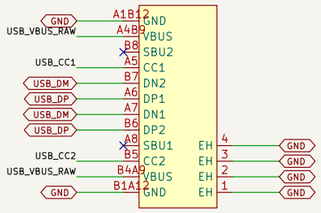
Title: Smart Blinds / integrated controller

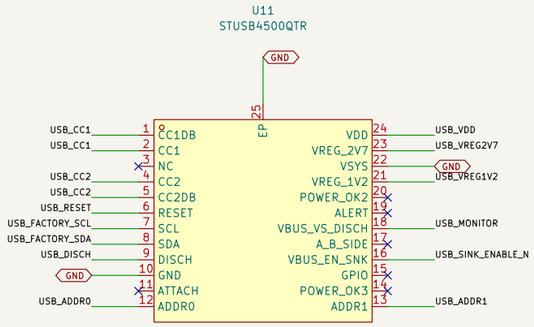
Size: A4	Date:	**Rev: STUDY**
KiCad E.D.A. 10.0.7		Id: 1/11

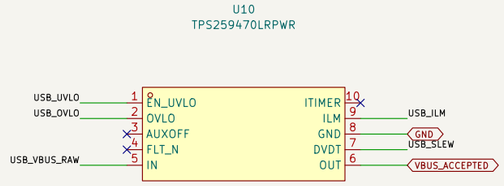
USB-C PD INPUT / CIRCUIT STUDY

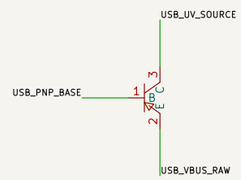
Factory NVM required: 5V/1.5A, 9V/2.5A, 15V/2.5A; POWER_ONLY_ABOVE_5V=1. Factory bus is separate from the MCU bus.

Source/load protection, discharge timing and startup remain unqualified. Charger input capacitor bank is external.

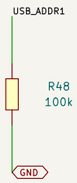


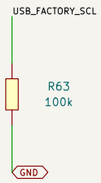


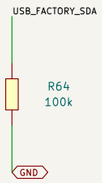


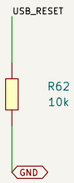




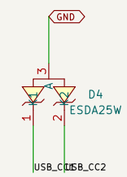


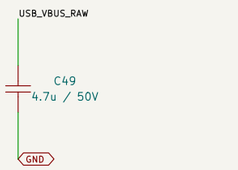










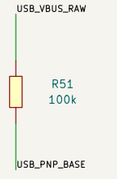


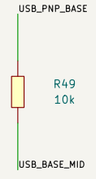


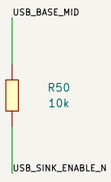


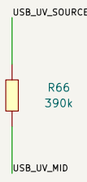


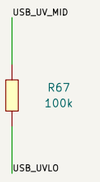


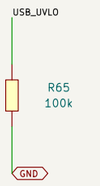




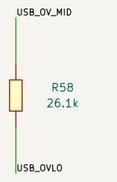


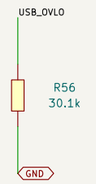


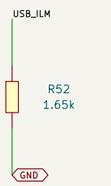




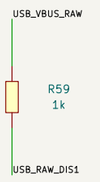


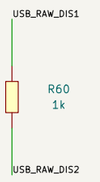




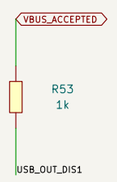


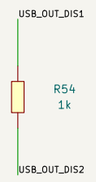


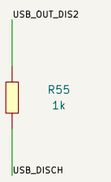












33-part unrouted capture. Native pin equivalence is not PD protocol, ESD/surge or fabrication approval.

Unrouted integrated circuit; not for fabrication

Sheet: /USB-C input/
File: 01-usb.kicad_sch

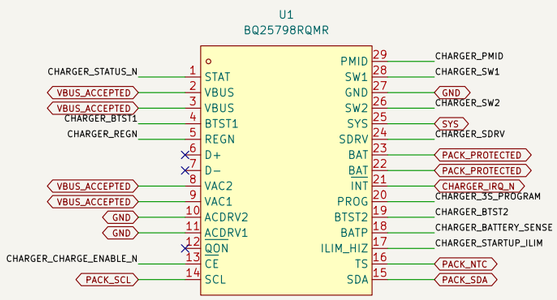
Title: Smart Blinds / USB-C input

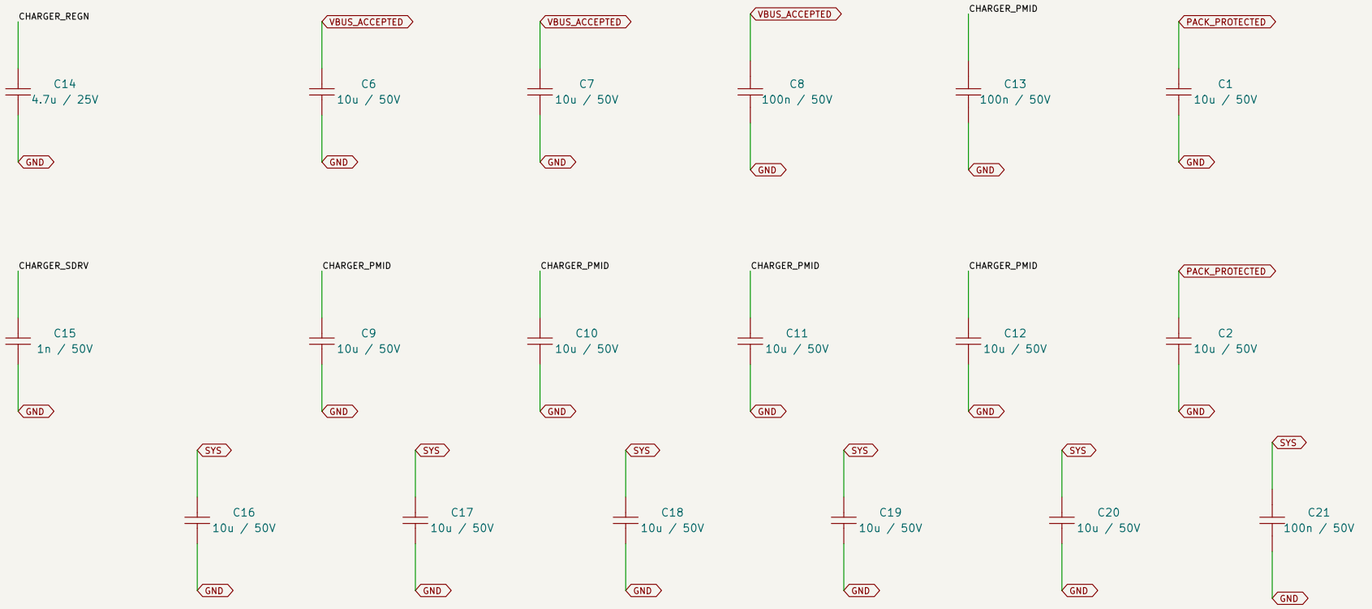
Size: A3	Date:	**Rev: STUDY**
KiCad E.D.A. 10.0.7		Id: 2/11

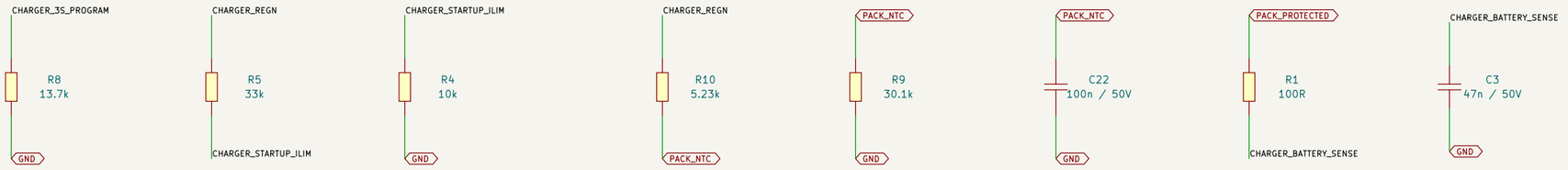
3S CHARGER / CIRCUIT STUDY

BQ25798: 3S, 750 kHz, 2.2 uH. Protected pack, USB-PD and logic regulator connect through named global nets.

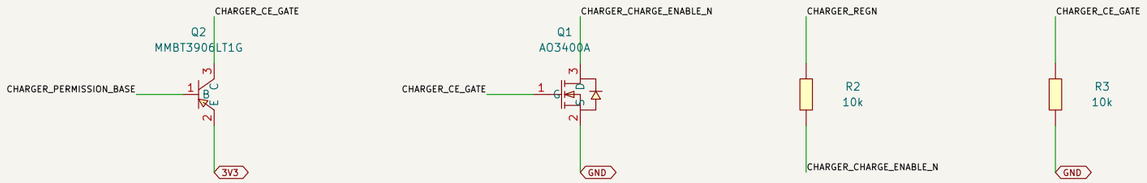
All capacitor values are nominal. DC bias, converter dynamics and thermal behavior require qualification.

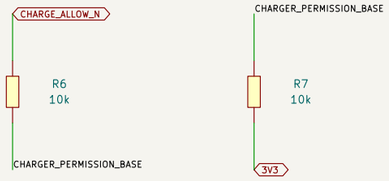






Active-LOW charging permission; absent 3V3 keeps the CE pull-down off.





NTC: independent 10k / B3435 thermistor on cells (external).
Host sets 40C warm suspend; independent pack protection required.
Default hard-hot is approximately 60C, not 45C.

Motor uses protected PACK, not SYS.
No placement, routing, physical charging or fabrication approval.

Integrated controller page. Native exported netlist checked against all PCB pins; unpowered and unrouted.

Unrouted integrated circuit; not for fabrication

Sheet: /3S charger/
File: 02-charger.kicad_sch

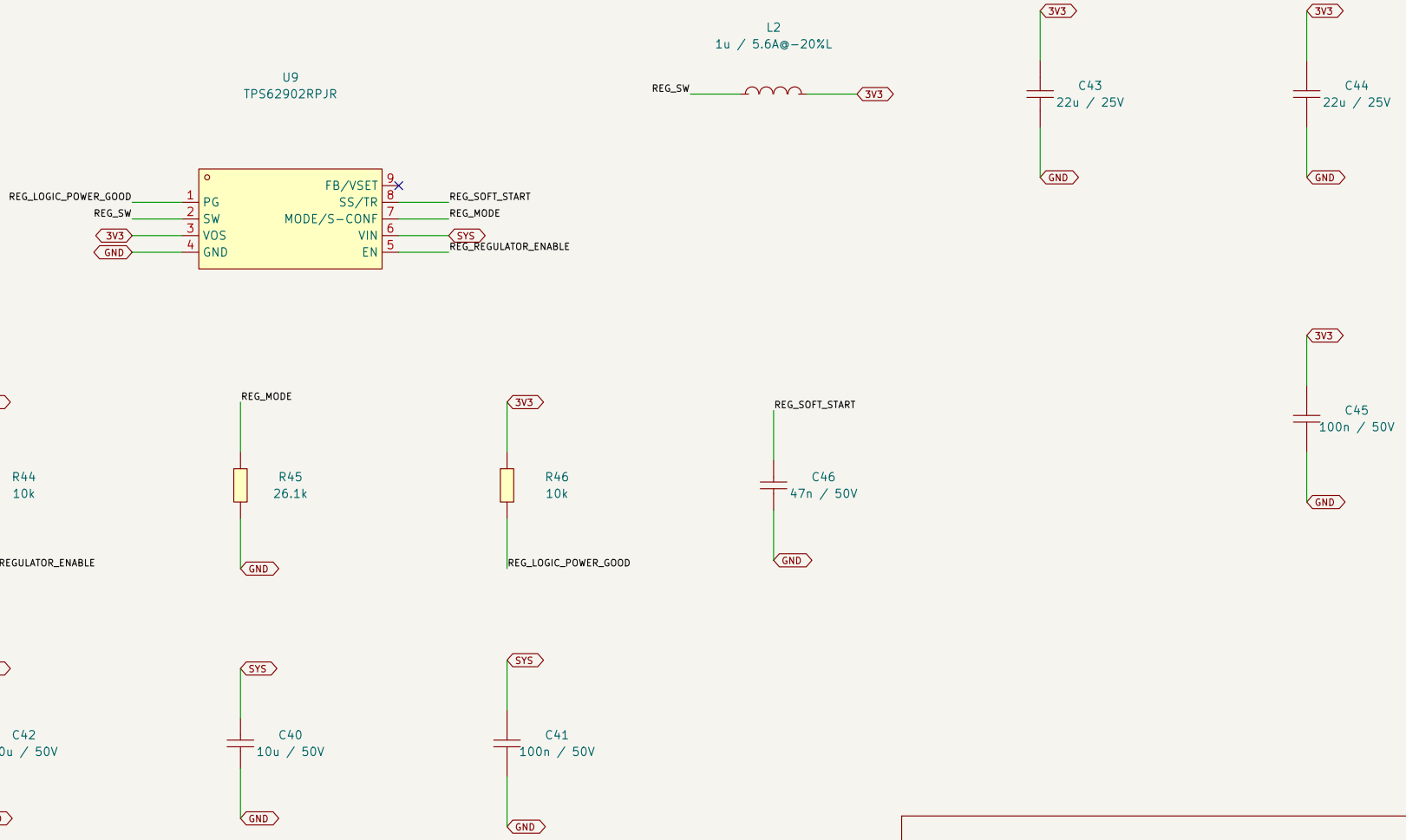
Title: Smart Blinds / 3S charger

Size: A3	Date:	**Rev: STUDY**
KiCad E.D.A. 10.0.7		Id: 3/11

3.3 V LOGIC SUPPLY / CIRCUIT STUDY

TPS62902: charger SYS 8.8–13.2V input; 1A load design target. Internal 3.3V divider (VSET intentionally open).

Capacitance is nominal. Typical DC-bias estimates and 65mV static reset margin do not prove load-transient behavior.



Integrated controller page. Native exported netlist checked against all PCB pins; unpowered and unrouted.

Unrouted integrated circuit; not for fabrication

Sheet: /Logic supply/
File: 03-logic.kicad_sch

Title: Smart Blinds / Logic supply

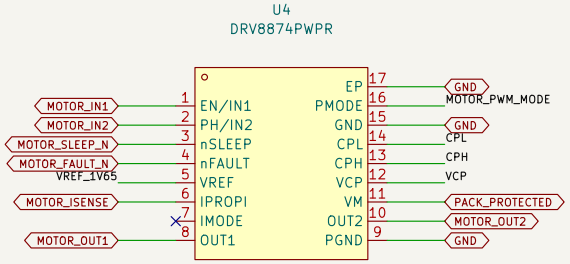
Size: A4	Date:	Rev: STUDY
KiCad E.D.A. 10.0.7		Id: 4/11

MOTOR DRIVER / CIRCUIT STUDY

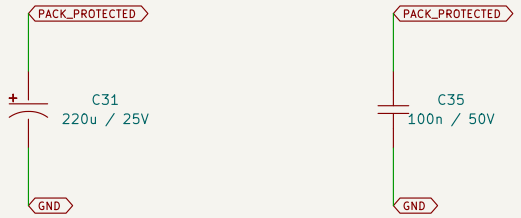
DRV8874; 9-12.6 V pack; 3.3 V logic. Global nets connect controller and protection sheets.

Default off via pull-downs. IMODE floats deliberately: fixed off-time / latched overcurrent fault.

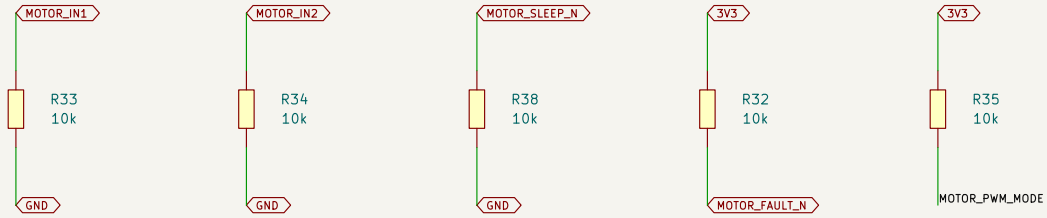
Control and status

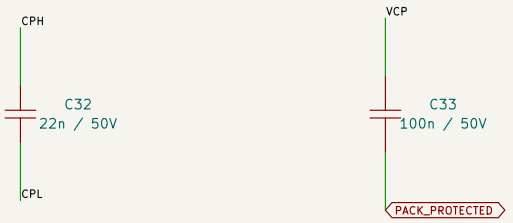


Supply and charge pump



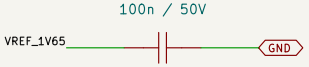
Default-off inputs / fault pull-up / PWM mode





Current threshold and measurement





Nominal current limit about 3 A; integrated upper estimate 3.418 A before transient overshoot. Motor data unmeasured.

Regenerative clamp appears on sheet 7. Thermal layout and powered qualification remain open.

Unrouted integrated circuit; not for fabrication

Sheet: /Motor bridge/
File: 04-motor.kicad_sch

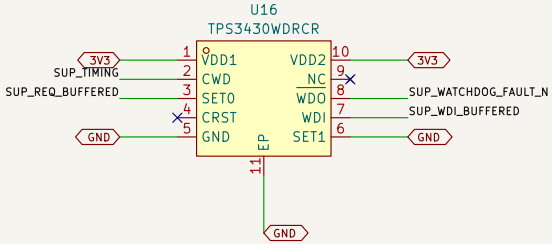
Title: Smart Blinds / Motor bridge

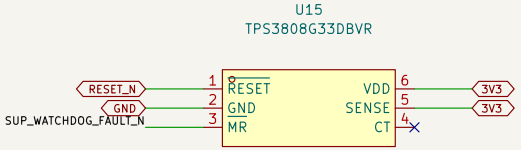
Size: A4	Date:	Rev: STUDY
KiCad E.D.A. 10.0.7		Id: 5/11

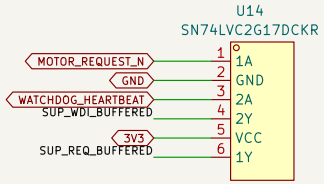
MOTOR SUPERVISION / CIRCUIT STUDY

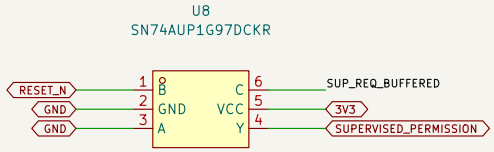
External 3.3 V supply. Reset and heartbeat boundary for ESP32-C6 and DRV8874.

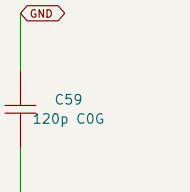
Request HIGH disables motor and watchdog. LOW requires healthy control-task falling edges.

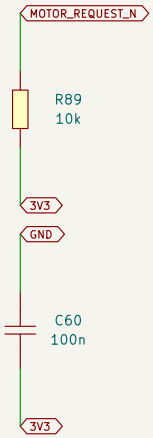


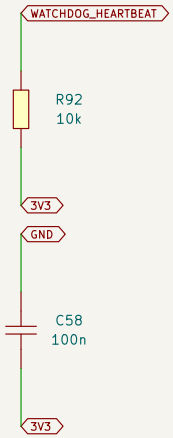


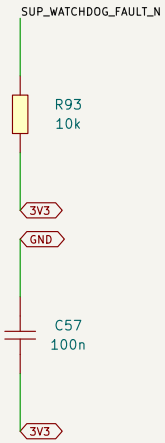














Upper window 57.74-71.35 ms; lower maximum 8.92 ms.

No EN capacitor. Reset cancels motion. Not for fabrication.

Unrouted integrated circuit; not for fabrication

Sheet: /Motor supervision/
File: 05-supervision.kicad_sch

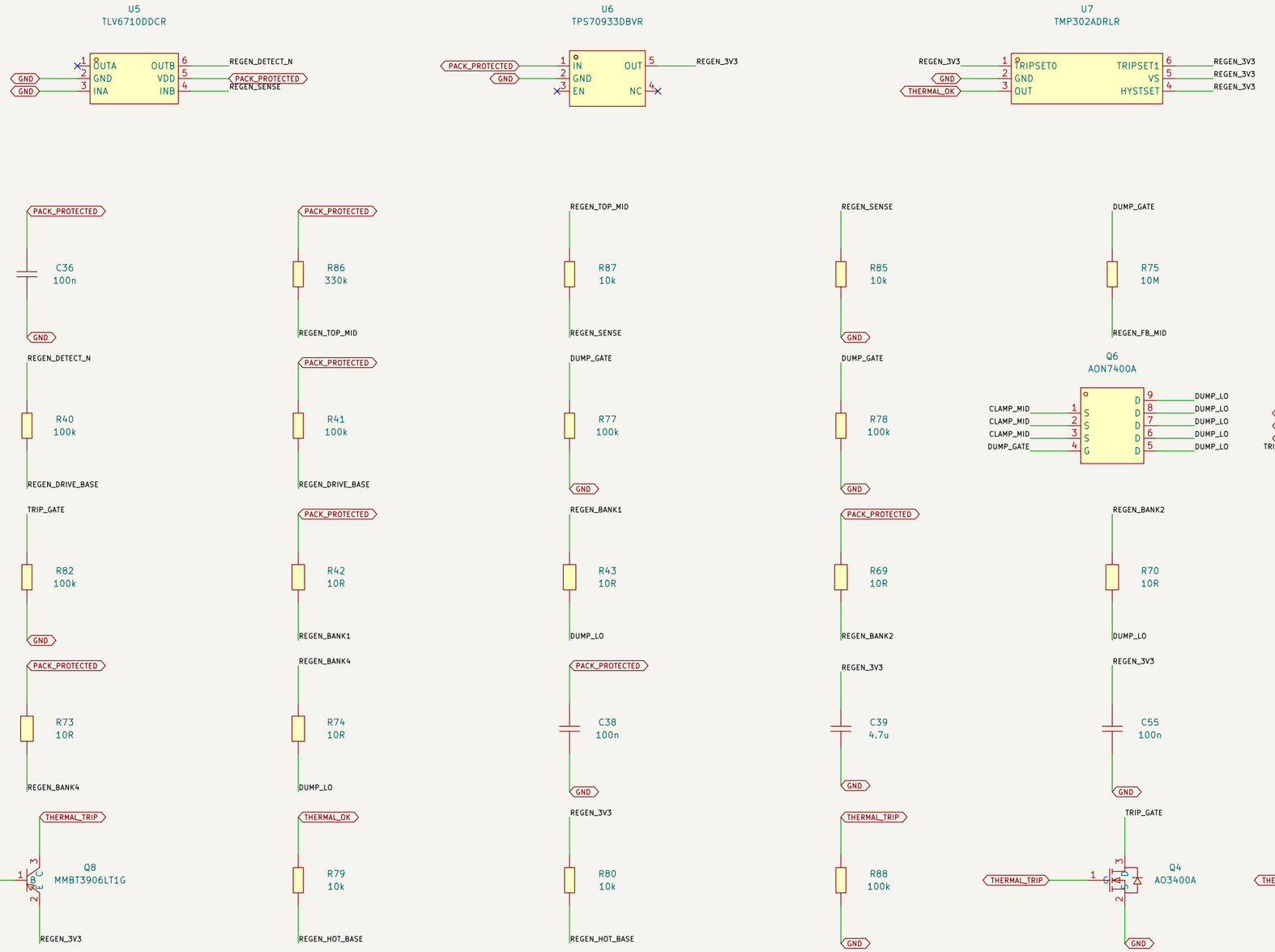
Title: Smart Blinds / Motor supervision

Size: A4	Date:	Rev: STUDY
KiCad E.D.A. 10.0.7		Id: 6/11

REGENERATIVE CLAMP / THERMAL LATCH STUDY

PACK−fed protection. 8 x 10R resistors: 2 series x 4 parallel = 5R, 65 C trip latches until PACK removal.

Pulse energy, switch delay, thermal coupling and sustained back−drive remain unqualified; simulation is conditional.



38−part unrouted capture. Native pin equivalence is not thermal, transient or fabrication approval.

Unrouted integrated circuit; not for fabrication

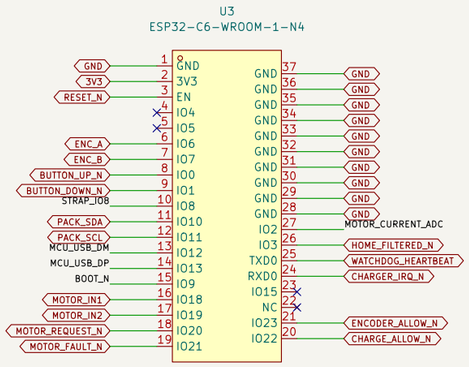
Sheet: /Regenerative clamp/
File: 06−regen.kicad_sch

Title: Smart Blinds / Regenerative clamp

Size: A3	Date:	Rev: STUDY
KiCad E.D.A. 10.0.7		Id: 7/11

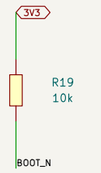
RADIO AND USB DATA / INTEGRATED CONTROLLER

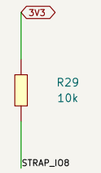
ESP32-C6. USB programming shares the USB-C port. Strap-only GPIO4/5/15 are intentionally left open.

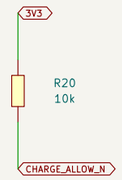


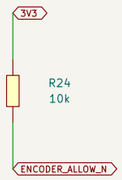


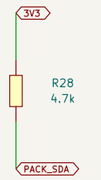


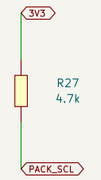


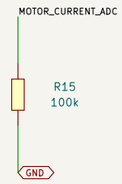


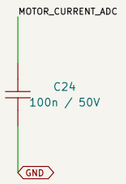












All pin connections independently exported and checked. Physical fit, routing and powered qualification remain open.

Unrouted integrated circuit; not for fabrication

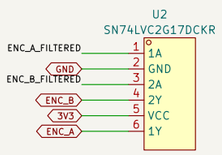
Sheet: /Radio and USB data/
File: 07-radio.kicad_sch

Title: Smart Blinds / Radio and USB data

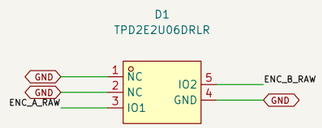
Size: A3	Date:	**Rev: STUDY**
KiCad E.D.A. 10.0.7		Id: 8/11

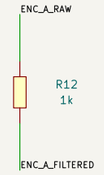
ENCODER INPUT / INTEGRATED CONTROLLER

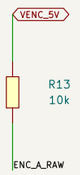
5 V tolerant Schmitt inputs. Connector pinout and supply must be checked on the owned motor before connection.

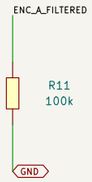


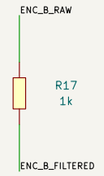


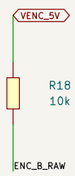


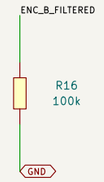


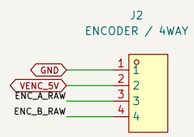












All pin connections independently exported and checked. Physical fit, routing and powered qualification remain open.

Unrouted integrated circuit; not for fabrication

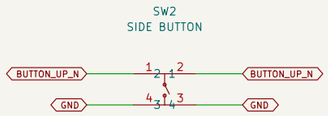
Sheet: /Encoder input/
File: 08-encoder-input.kicad_sch

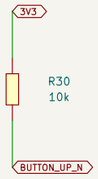
Title: Smart Blinds / Encoder input

Size: A3	Date:	**Rev: STUDY**
KiCad E.D.A. 10.0.7		Id: 9/11

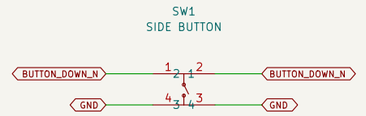
BUTTONS AND HOME INPUT / INTEGRATED CONTROLLER

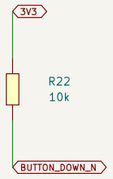
Home connector: pin1 GND, pin2 reserved/open, pin3 dry contact. Switch terminal pairing needs sample continuity check.



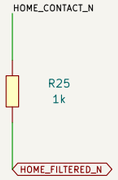


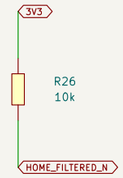




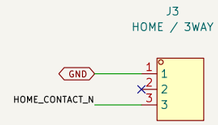


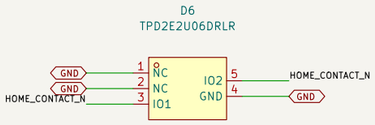












All pin connections independently exported and checked. Physical fit, routing and powered qualification remain open.

Unrouted integrated circuit; not for fabrication

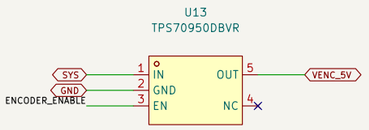
Sheet: /Buttons and home input/
File: 09-local-input.kicad_sch

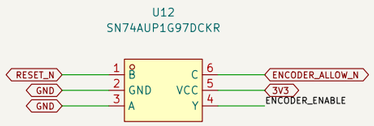
Title: Smart Blinds / Buttons and home input

Size: A3	Date:	**Rev: STUDY**
KiCad E.D.A. 10.0.7		Id: 10/11

POWER AND MOTOR PERMISSION / INTEGRATED CONTROLLER

Encoder gate: RESET_N AND NOT ALLOW_N. Final motor gate: supervision AND thermal permission. No raw cell rails here.

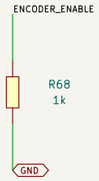


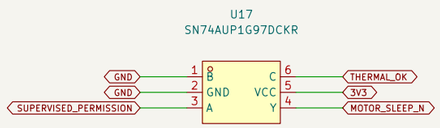






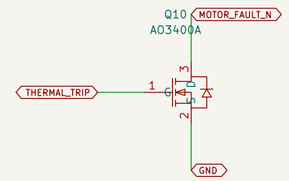


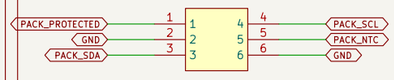
















All pin connections independently exported and checked. Physical fit, routing and powered qualification remain open.

Unrouted integrated circuit; not for fabrication

Sheet: /Power and motor permission/
File: 10-interfaces.kicad_sch

Title: Smart Blinds / Power and permission

Size: A3	Date:	**Rev: STUDY**
KiCad E.D.A. 10.0.7		Id: 11/11